

Overview

Part 03 — 🧠 Natural Language Processing · 03-nlp/README.md

🧠 Module 03 — Natural Language Processing

Where you are: Stop 3 of 16. **Time:** ~25 hours · **Prereq:** Module 02 [02-deep-learning].

This is the hinge of the whole track. NLP is where representation learning met sequences and produced the transformer — the architecture that every model in Modules 04–11 is built from. Do not skim this module to get to LLMs faster. **This module is how LLMs work**; Module 04 is what happens when you scale it.

Learning objectives

1. Explain the full text pipeline: raw text → tokens → embeddings → contextual representations → task output.
2. Implement **BPE tokenization** and explain why tokenization causes half of the weird LLM behaviours you'll see.
3. Derive scaled dot-product attention and implement multi-head self-attention from scratch.
4. Draw a transformer block from memory, including the position of every LayerNorm and residual.
5. Distinguish encoder-only (BERT), decoder-only (GPT), and encoder–decoder (T5) models and pick correctly.
6. Explain positional encoding — sinusoidal, learned, and **RoPE** — and why it determines context length.

Suggested path

#	Do this	File	Time
1	The pipeline and its history	notes/01-text-to-vectors.md	4h
2	The transformer, in full	notes/02-transformers.md	5h
3	Build a BPE tokenizer	code/bpe_tokenizer.py	3h
4	Build attention	code/attention_from_scratch.py	4h
5	Slides	slides/	1h
6	Lab	lab/EXERCISES.md	6h
7	Papers (read #1 twice)	papers/PAPERS.md	5h

The 14 terms you must own

token · BPE · vocabulary · embedding · positional encoding · RoPE · self-attention · query/key/value · multi-head · causal mask · KV cache · encoder-only · decoder-only · perplexity

Exit check

You can implement a working single-head and multi-head attention, apply a causal mask correctly, explain why `strawberry` has three r's but the model can't count them, and sketch a transformer block on a napkin.